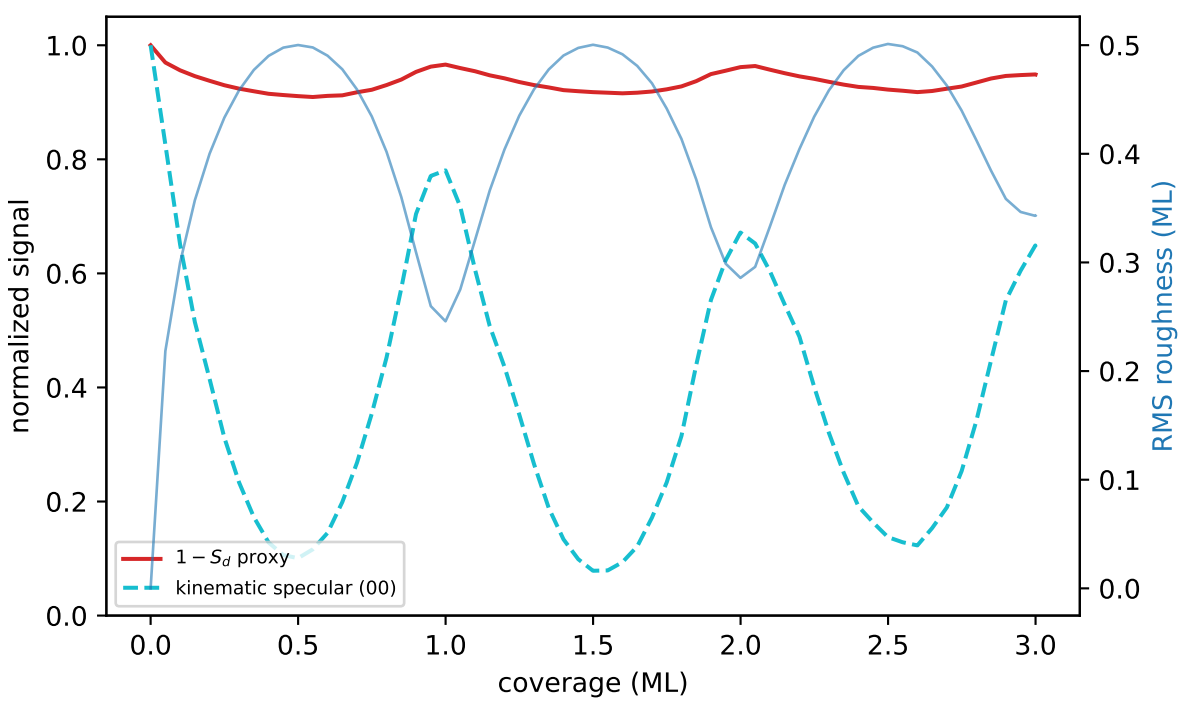
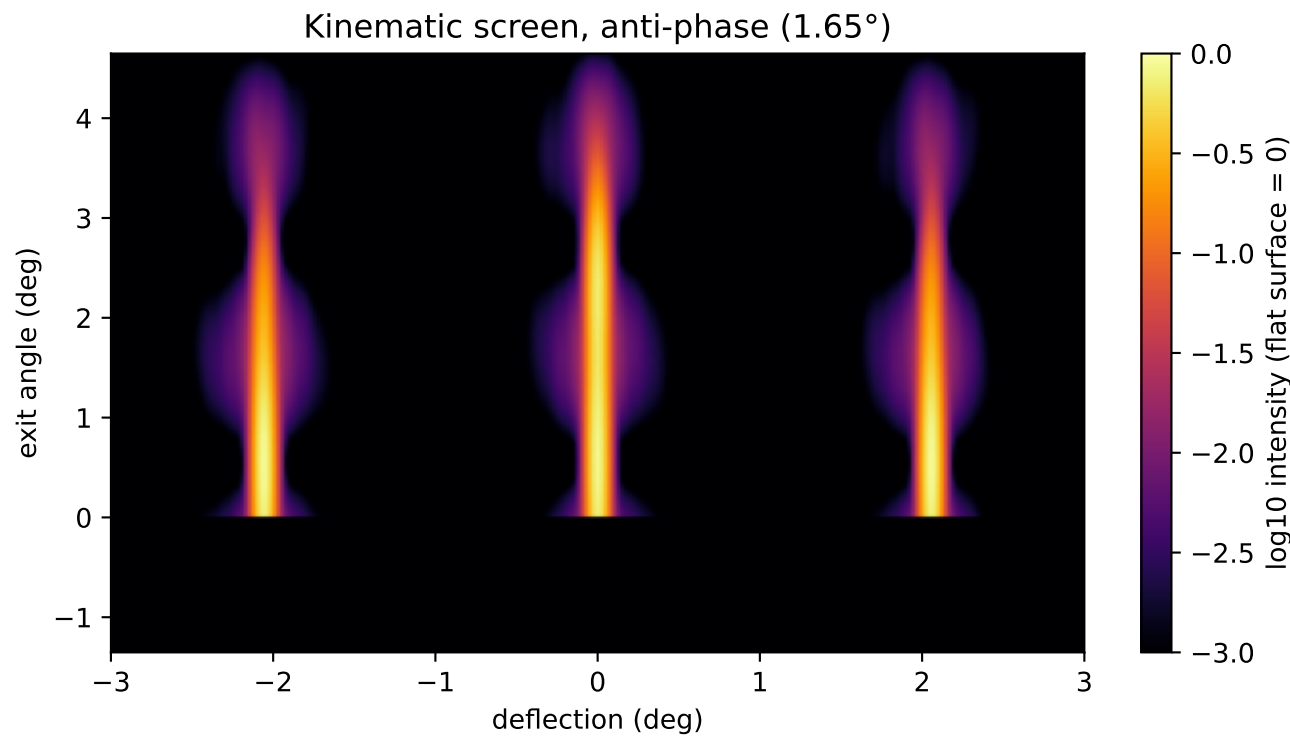
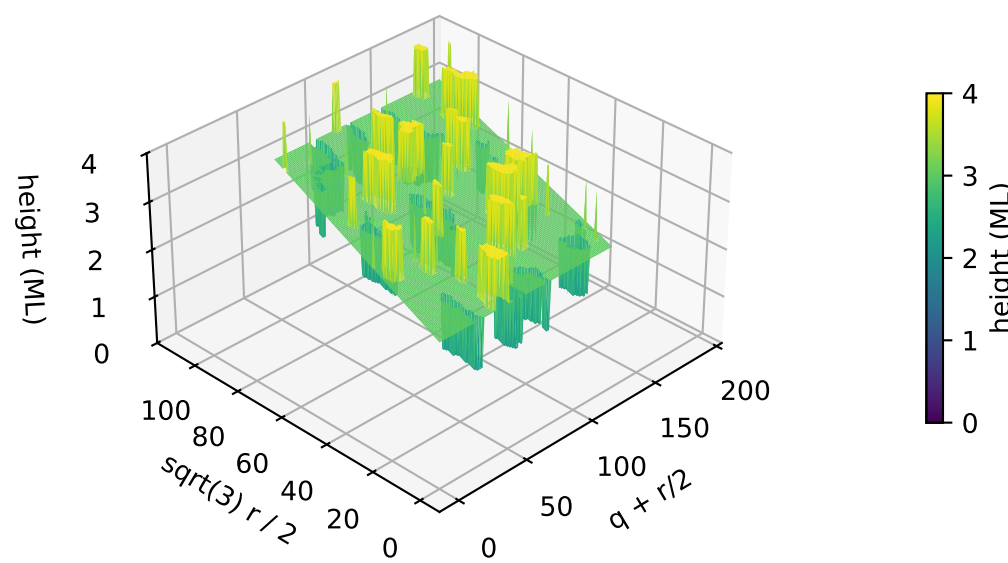


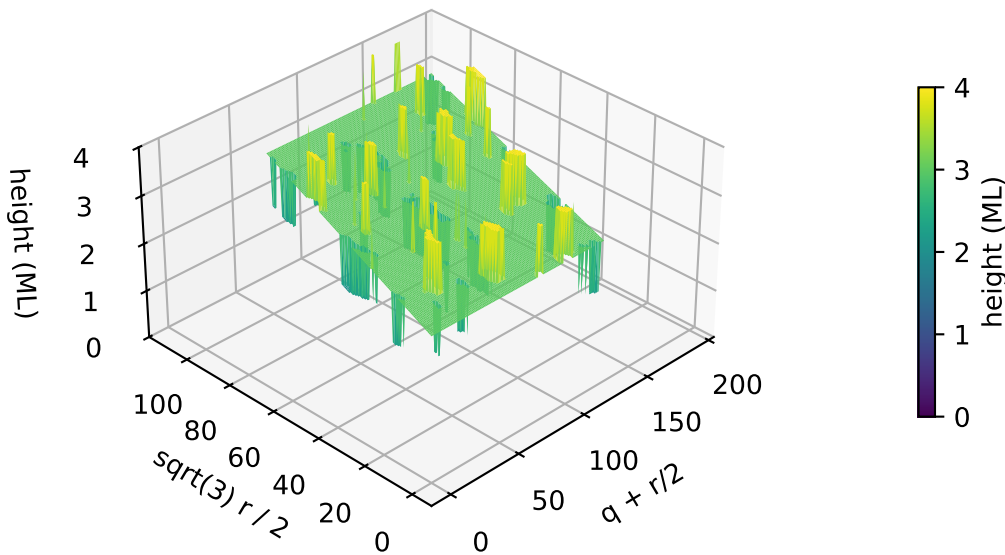
Run 1 of 6 — Layer-by-layer growth (GaN paper parameters)
Ga/N = 0.82 paper preset, unmodified; predicted growth rate 0.2282 ML/s
Final surface, 3.00 ML



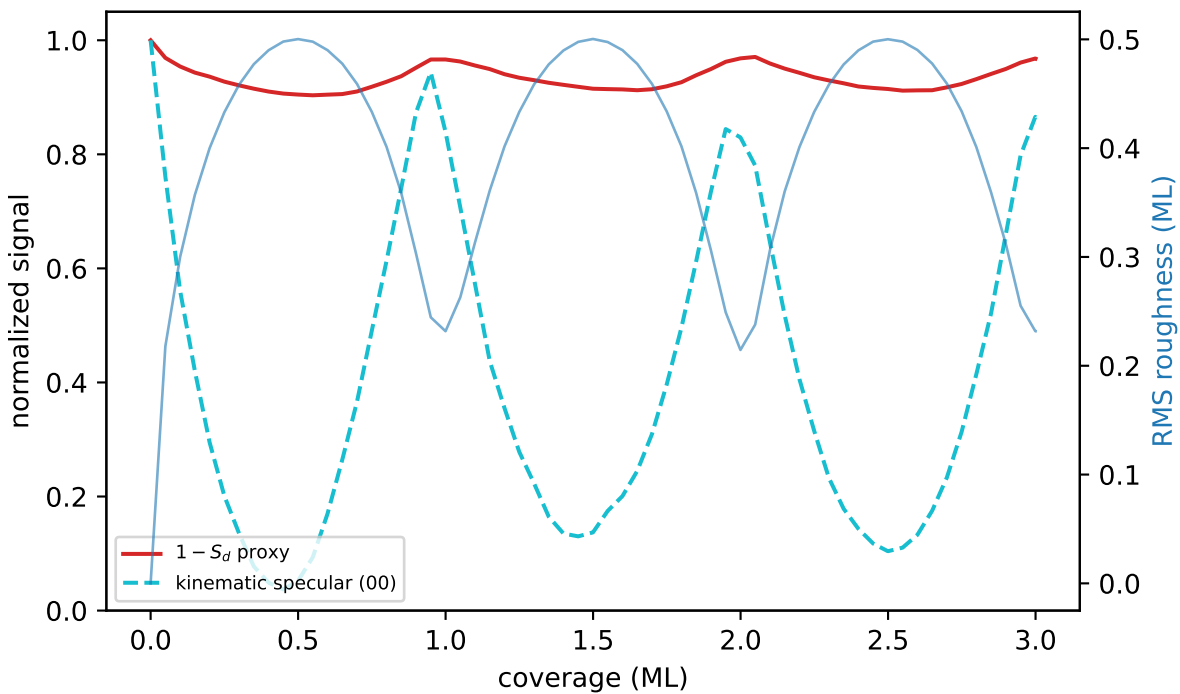
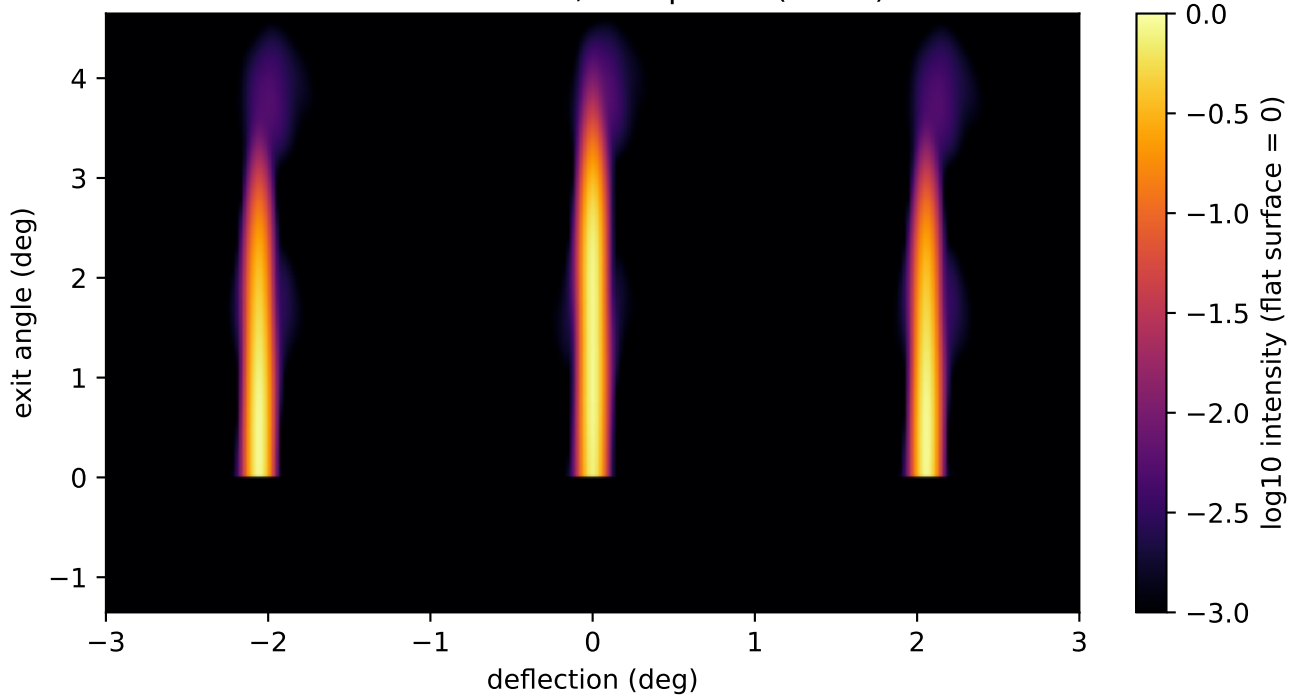
run number	1 of 6
preset key	gan-paper-082
lattice	128x128 (16384 sites)
target coverage	3.00 ML
temperature	1003.1 K
deposition flux	0.2450 ML/s
attempt frequency	1e+13 Hz
E_diff	1.6188 eV
E_bond	0.3050 eV
E_step	0.0659 eV
E_des	2.3813 eV
max hop distance	5
seed	7
frames stored	62
simulated time	12.521 s
deposited / hops / desorbed	50229 / 7521761 / 1077
final RMS roughness	0.3430 ML
island density peak / final	0.00330 / 0.00214 per site
proxy period	1.033 ML
proxy oscillatory	True
specular final / flat	0.6454
beam	15 keV, 1.648°, 4.0 nm coherence
satellite ratio (box geometry)	0.157
wall time	10.0 s

Run 2 of 6 — Enhanced interlayer smoothing (no Ehrlich-Schwoebel barrier)
derived from the Ga/N = 0.82 paper preset; modified: E_step = 0.0000 eV (paper 0.0659 eV)

Final surface, 3.00 ML



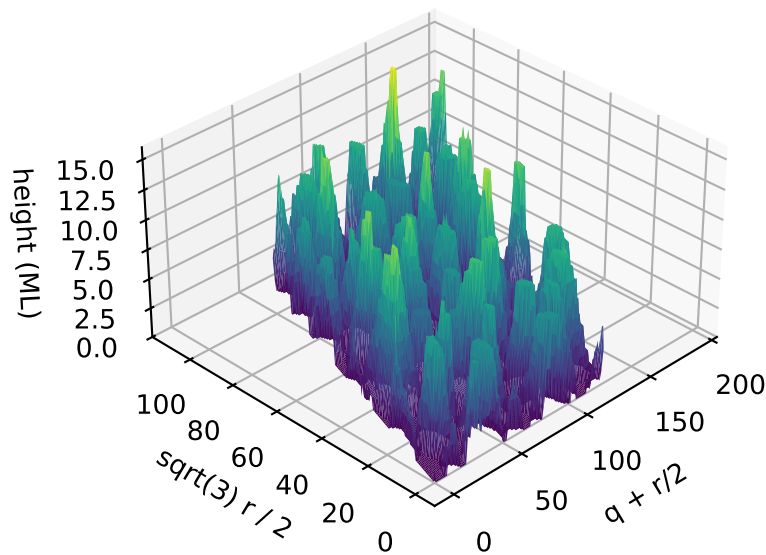
Kinematic screen, anti-phase (1.65°)



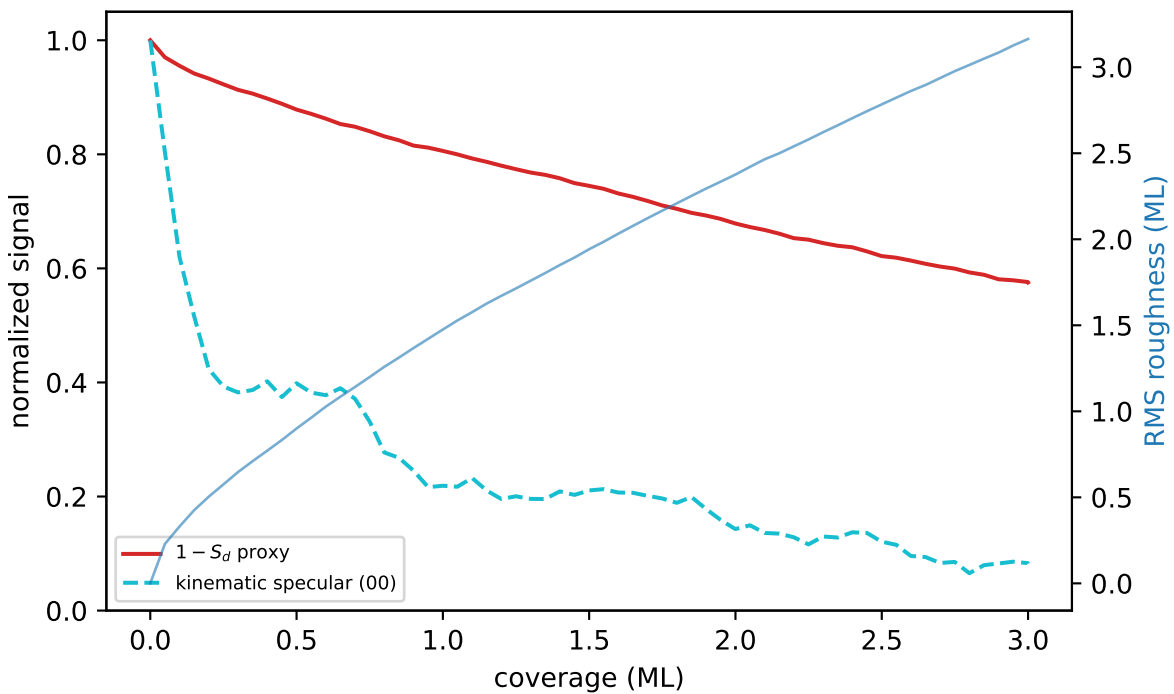
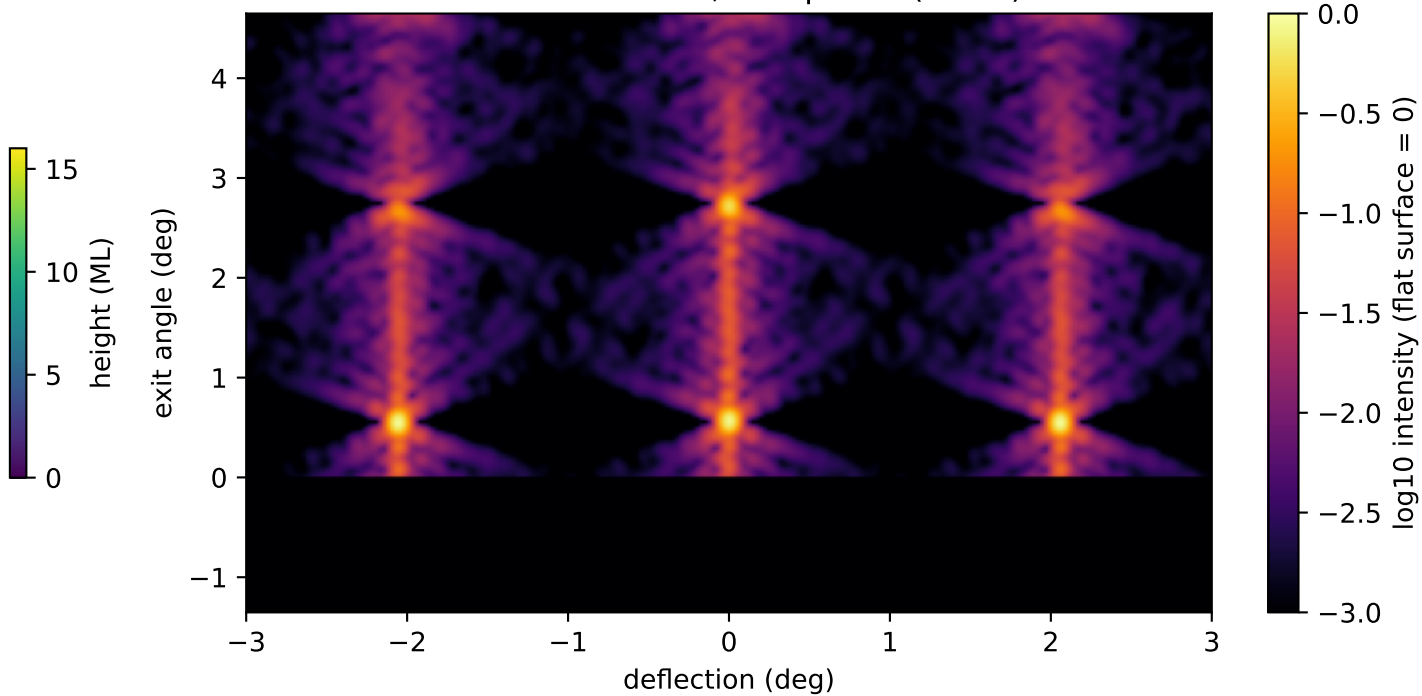
run number	2 of 6
preset key	island-growth
lattice	128x128 (16384 sites)
target coverage	3.00 ML
temperature	1003.1 K
deposition flux	0.2450 ML/s
attempt frequency	1e+13 Hz
E_diff	1.6188 eV
E_bond	0.3050 eV
E_step	0.0000 eV
E_des	2.3813 eV
max hop distance	5
seed	7
frames stored	62
simulated time	12.542 s
deposited / hops / desorbed	50132 / 6574028 / 980
final RMS roughness	0.2318 ML
island density peak / final	0.00342 / 0.00201 per site
proxy period	1.033 ML
proxy oscillatory	True
specular final / flat	0.8652
beam	15 keV, 1.648°, 4.0 nm coherence
satellite ratio (box geometry)	0.157
wall time	7.8 s

Run 3 of 6 — Mounding from a strong Ehrlich-Schwoebel barrier
derived from the Ga/N = 0.82 paper preset; modified: E_step = 0.2500 eV (paper 0.0659 eV)

Final surface, 3.00 ML

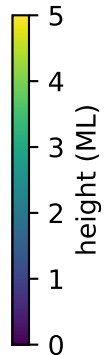
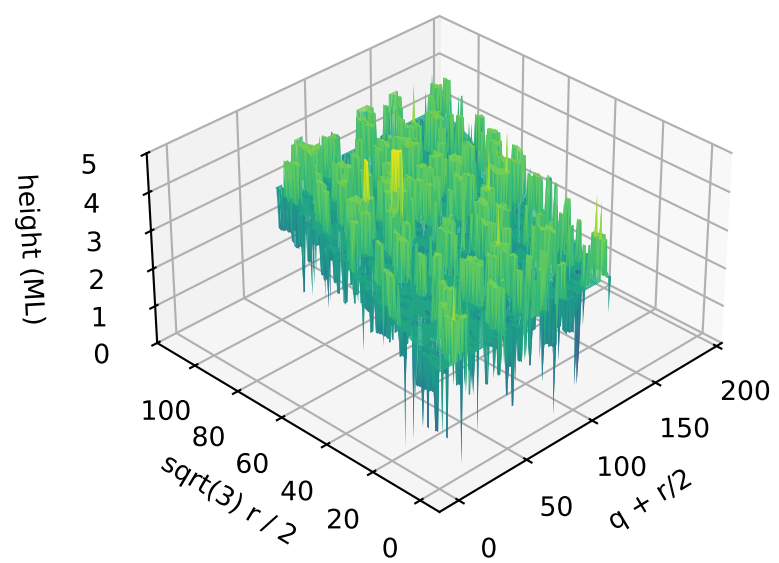


Kinematic screen, anti-phase (1.65°)

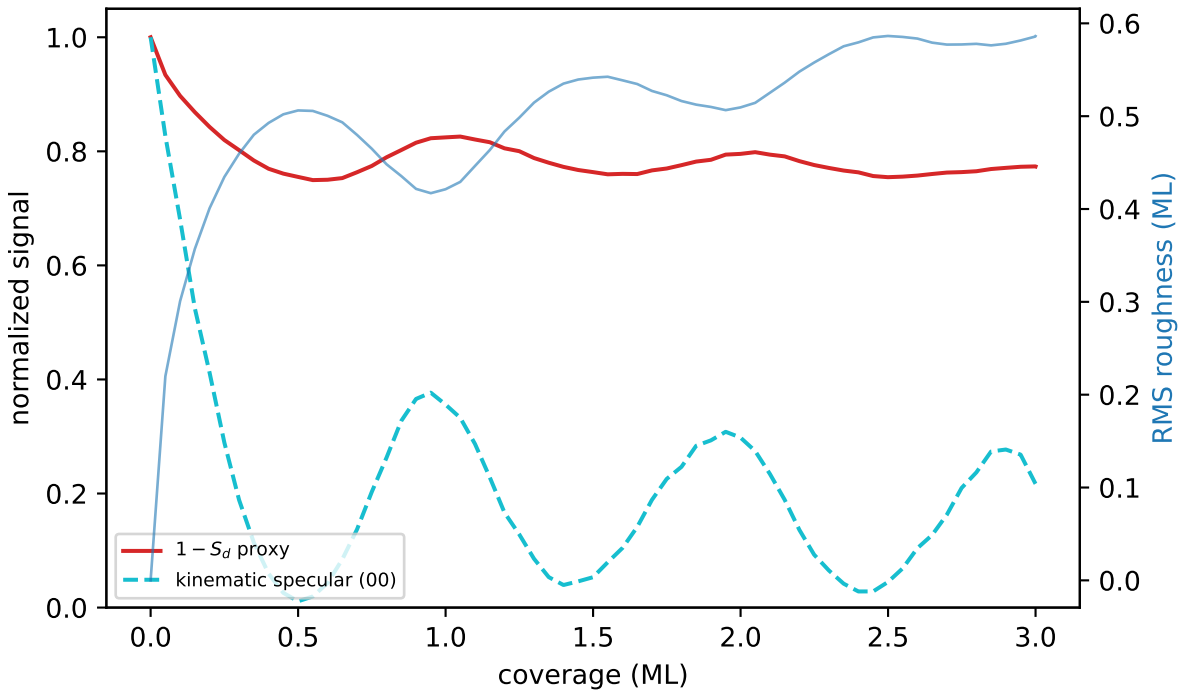
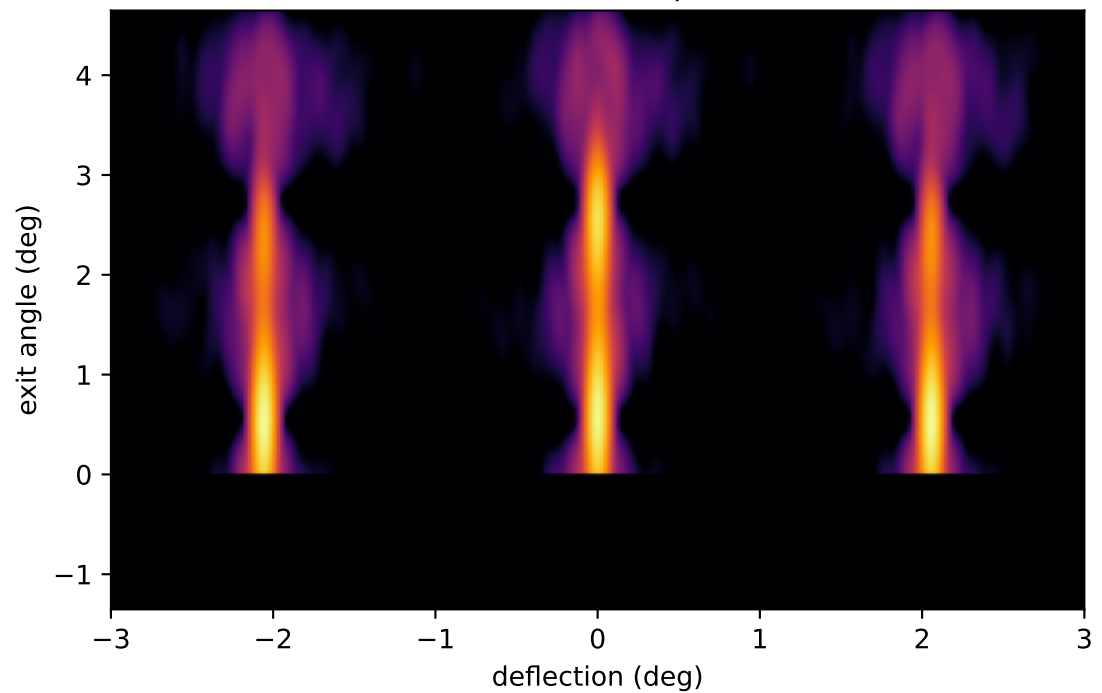


run number	3 of 6
preset key	step-barrier-mounding
lattice	128x128 (16384 sites)
target coverage	3.00 ML
temperature	1003.1 K
deposition flux	0.2450 ML/s
attempt frequency	1e+13 Hz
E_diff	1.6188 eV
E_bond	0.3050 eV
E_step	0.2500 eV
E_des	2.3813 eV
max hop distance	5
seed	7
frames stored	62
simulated time	12.549 s
deposited / hops / desorbed	50252 / 5963518 / 1100
final RMS roughness	3.1643 ML
island density peak / final	0.00323 / 0.00214 per site
proxy period	n/a
proxy oscillatory	False
specular final / flat	0.0822
beam	15 keV, 1.648°, 4.0 nm coherence
satellite ratio (box geometry)	0.157
wall time	22.7 s

Run 4 of 6 — Low temperature: reduced mobility, dense nucleation
derived from the Ga/N = 0.82 paper preset; modified: T = 850.0 K (paper 1003.1 K)
Final surface, 3.00 ML

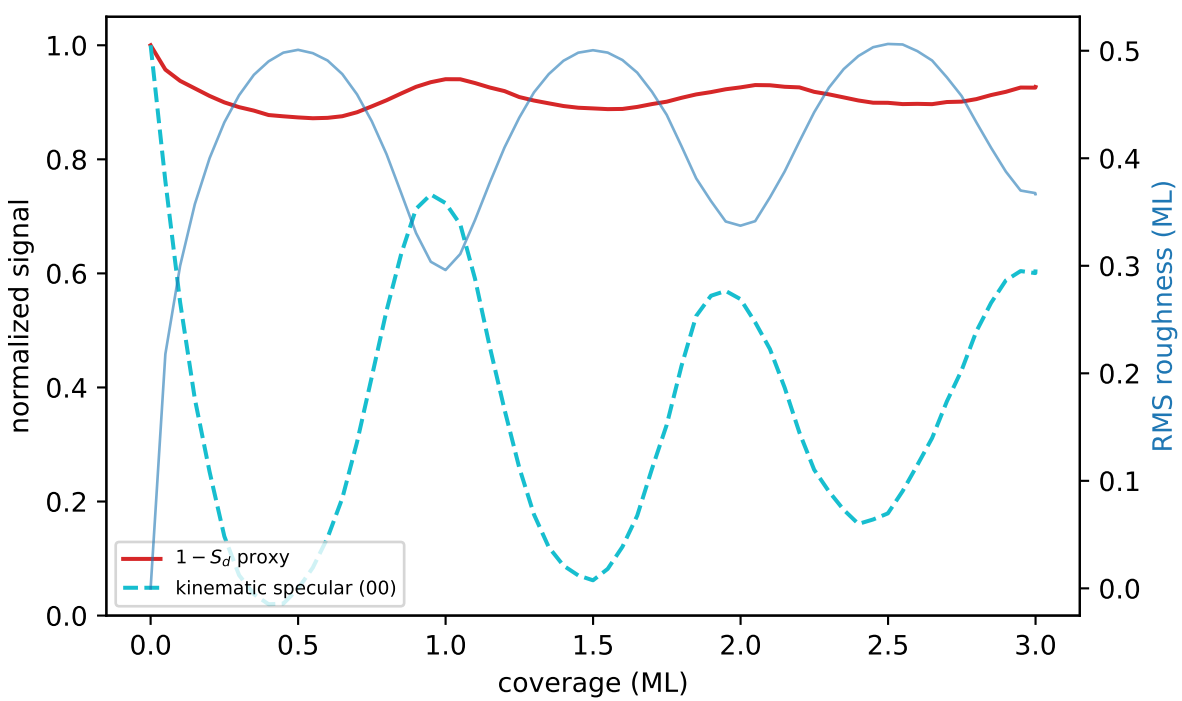
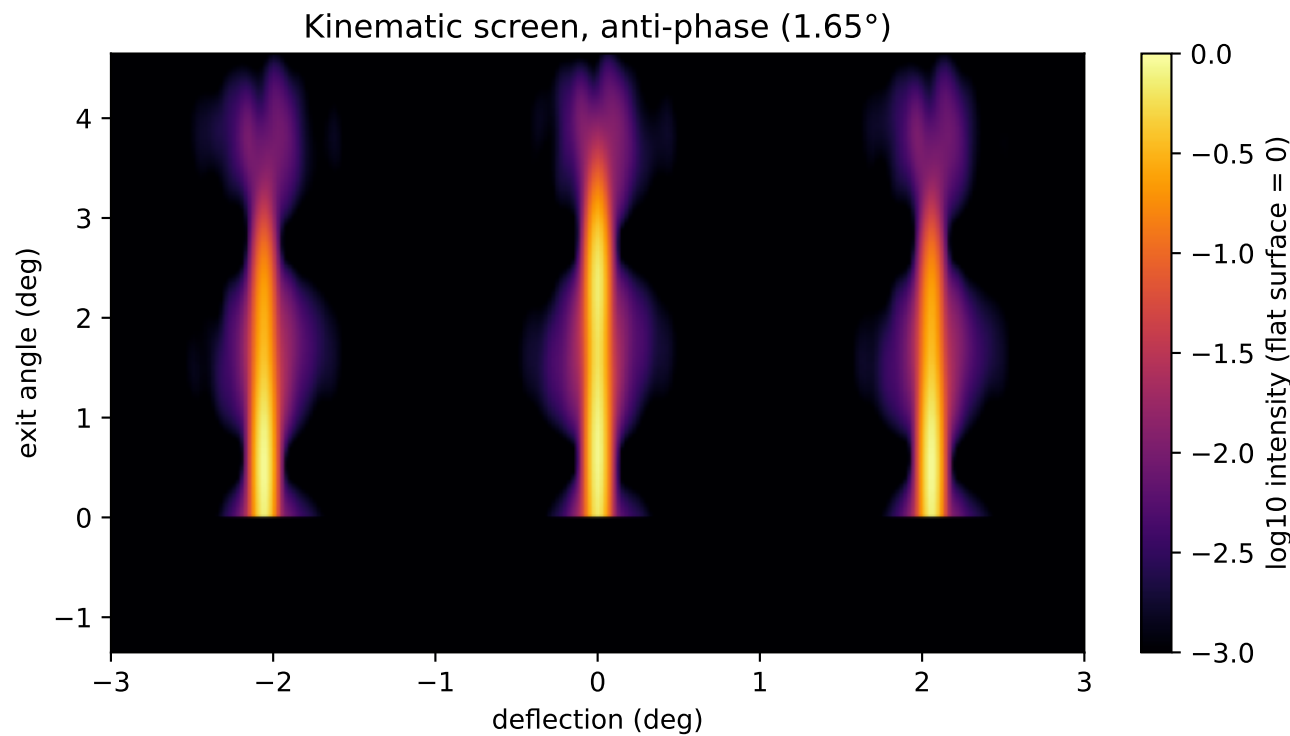
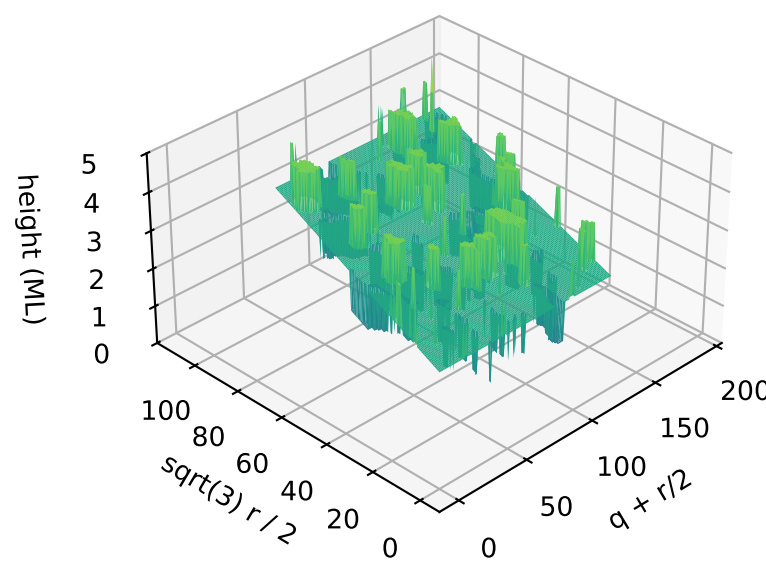


Kinematic screen, anti-phase (1.65°)



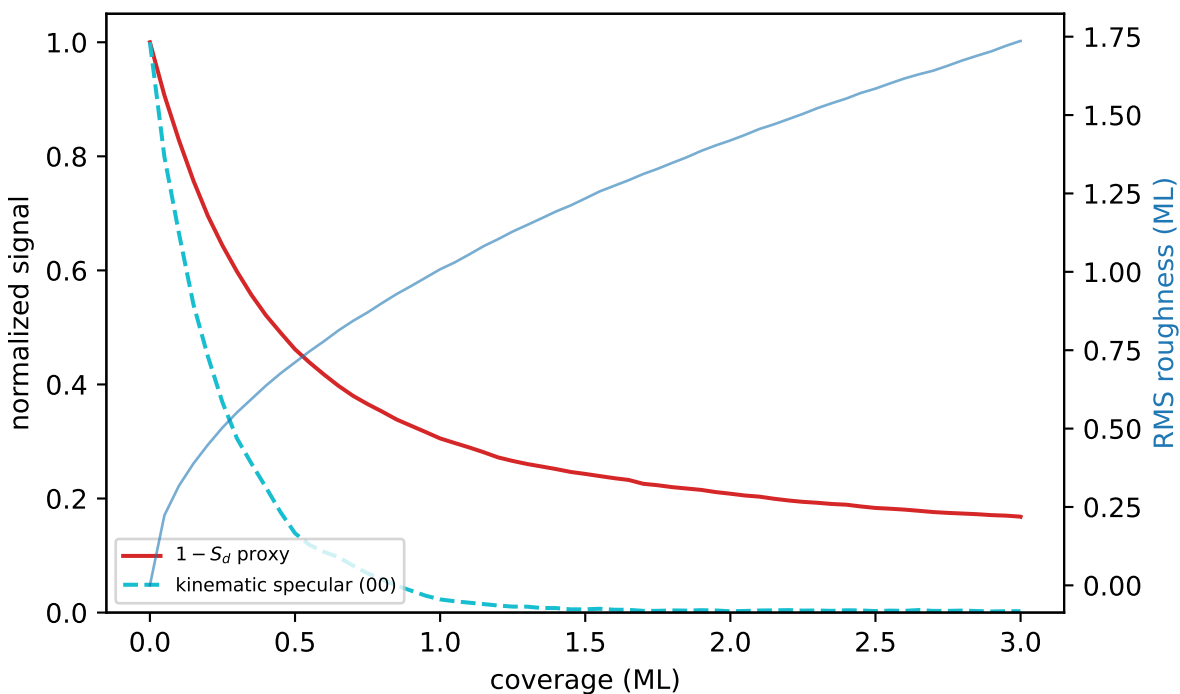
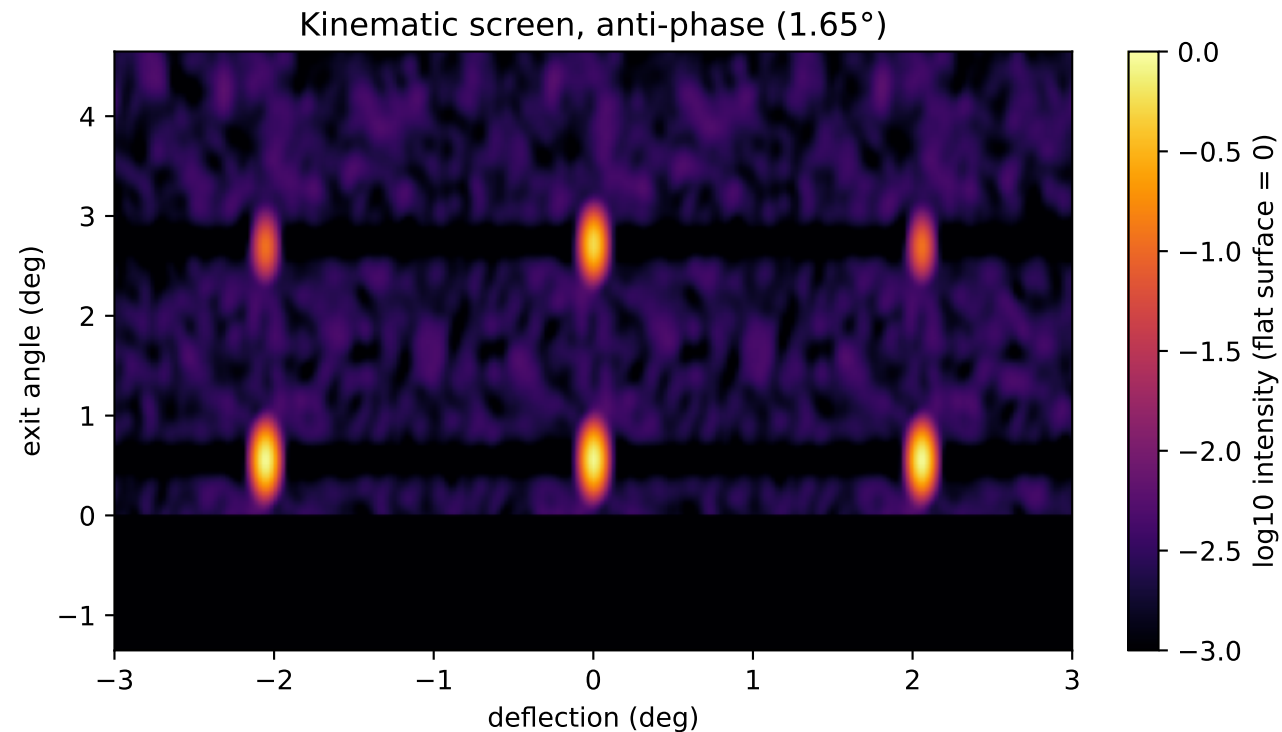
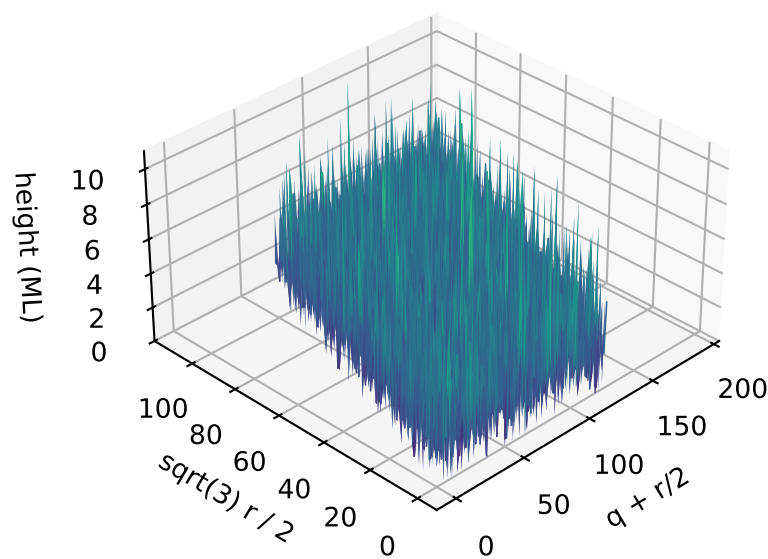
run number	4 of 6
preset key	too-cold
lattice	128x128 (16384 sites)
target coverage	3.00 ML
temperature	850.0 K
deposition flux	0.2450 ML/s
attempt frequency	1e+13 Hz
E_diff	1.6188 eV
E_bond	0.3050 eV
E_step	0.0659 eV
E_des	2.3813 eV
max hop distance	5
seed	7
frames stored	62
simulated time	12.311 s
deposited / hops / desorbed	49169 / 747947 / 17
final RMS roughness	0.5862 ML
island density peak / final	0.01544 / 0.00745 per site
proxy period	1.033 ML
proxy oscillatory	True
specular final / flat	0.2170
beam	15 keV, 1.648°, 4.0 nm coherence
satellite ratio (box geometry)	0.157
wall time	3.9 s

Run 5 of 6 — High flux: 3x effective Ga flux
derived from the Ga/N = 0.82 paper preset; modified: effective Ga flux = 0.7349 ML/s (paper 0.2450 ML/s)
Final surface, 3.00 ML



run number	5 of 6
preset key	too-fast
lattice	128x128 (16384 sites)
target coverage	3.00 ML
temperature	1003.1 K
deposition flux	0.7349 ML/s
attempt frequency	1e+13 Hz
E _{diff}	1.6188 eV
E _{bond}	0.3050 eV
E _{step}	0.0659 eV
E _{des}	2.3813 eV
max hop distance	5
seed	7
frames stored	62
simulated time	4.131 s
deposited / hops / desorbed	49693 / 3749074 / 541
final RMS roughness	0.3666 ML
island density peak / final	0.00647 / 0.00293 per site
proxy period	1.033 ML
proxy oscillatory	True
specular final / flat	0.6069
beam	15 keV, 1.648°, 4.0 nm coherence
satellite ratio (box geometry)	0.157
wall time	7.2 s

Run 6 of 6 — No diffusion: pure random deposition
derived from the Ga/N = 0.82 paper preset; modified: attempt frequency = 0 Hz (paper 1e+13 Hz)
Final surface, 3.00 ML



run number	6 of 6
preset key	no-diffusion
lattice	128x128 (16384 sites)
target coverage	3.00 ML
temperature	1003.1 K
deposition flux	0.2450 ML/s
attempt frequency	0 Hz
E_diff	1.6188 eV
E_bond	0.3050 eV
E_step	0.0659 eV
E_des	2.3813 eV
max hop distance	5
seed	7
frames stored	62
simulated time	12.190 s
deposited / hops / desorbed	49152 / 0 / 0
final RMS roughness	1.7363 ML
island density peak / final	0.09839 / 0.06891 per site
proxy period	n/a
proxy oscillatory	False
specular final / flat	0.0024
beam	15 keV, 1.648°, 4.0 nm coherence
satellite ratio (box geometry)	0.157
wall time	0.7 s